Active Foveated Vision with Joint Policy Learning for Efficient RL Agents

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***Abstract*—In this project, we explore the training of reinforcement learning agents equipped with foveated vision to play the game of *Pong*. Using the SUGARL framework, we jointly learn both the motor policy and the sensory (vision) policy. We compare several architectural variants and vision strategies, with a focus on evaluating whether these foveated agents can achieve improved computational efficiency without sacrificing performance. Our experiments highlight the trade-offs between different designs in terms of average episodic rewards, inference time, and computational cost. While some architectures show promise in reducing the number of operations, none have yet matched the performance of traditional full-vision agents. These results suggest that achieving efficiency gains in such settings may require further improvements, particularly in memory mechanisms and architectural design.**

I. Project Summary

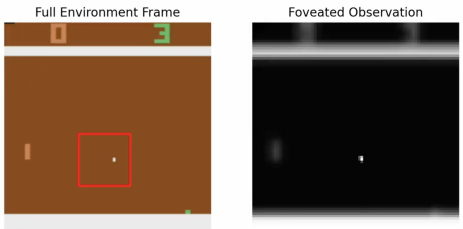


Figure 1: Our foveated vision setup

Humans and most animals do not process their entire visual field uniformly—doing so would be too computationally costly. Instead, they rely on foveated vision, processing only a small, high-resolution region at a time while the periphery remains low-resolution. This project investigates whether a similar strategy can benefit reinforcement learning (RL) agents. We consider a simplified setting using Gym's Pong, where the agent observes a low-resolution view of the environment along with a movable, high-resolution fovea. The environment is chosen for its simplicity and the interpretability of the vision policy—one might reasonably expect it to track the ball and sometimes check the location of the paddle. A key component of the project is the introduction of a learnable vision policy, which decides where the fovea should focus. However, prior work [1] has shown that naively combining a motor policy and a vision policy into a single global policy and training it with standard RL techniques often results in learning failure, since both components are interdependent and need to co-adapt during training. To address this, we use SUGARL [1], a method where a good vision policy is one that preserves enough information across consecutive frames to infer the motor action. Our goal is to implement and compare different foveated vision strategies—including SUGARL, handcrafted vision tracker and full vision and evaluate their performance, learned behavior, and computational efficiency. We hope that this toy setting provides insights applicable to real-world systems, such as lightweight robots or drones, where processing power is limited and foveated vision could offer a critical efficiency advantage or be able to work with smaller field of views.

II. Related Work

Our project was primarily inspired by two recent papers addressing active vision in reinforcement learning settings.

- In *Foveal Vision Reduces Neural Resources in Agent-Based Game Learning* [2], Chen et al. demonstrated that combining a motor policy with a foveated vision policy can lead to substantial computational savings—specifically in terms of the number of neurons and operations—when training agents to play *Pong*. However, their vision policy was guided by a handcrafted objective: tracking the ball. While effective in a simple environment like *Pong*, this approach does not generalize to more complex settings, where defining such an explicit objective is either infeasible or ill-defined.
- In contrast, *Active Vision Reinforcement Learning under Limited Visual Observability* [1] introduces SUGARL, a framework for jointly learning both the motor and vision policies without requiring a handcrafted objective for visual attention. This makes it more broadly applicable to complex environments where it is difficult—or even impossible—to specify what the agent should attend to a priori.

III. Method

A. Overview

We conducted a series of experiments to explore the capabilities and limitations of the SUGARL framework. Our primary goals were to understand what sensory and motor policies foveated agents are able to learn, assess the impact of architectural changes on computational efficiency, and

evaluate alternative design choices. The specific questions addressed are detailed in the next section.

B. Defining rewards

A game of *Pong* ends when a player reaches a score of 21. If we define the agent's reward as the difference between its score and the opponent's, then the final cumulative reward can range from –21 (losing 0–21) to +21 (winning 21–0).

C. Experimental Setup

In all experiments, the fovea is defined as an $F \times F$ square region, and the periphery as a $P \times P$ square. We denote a configuration as $[F, P]$, where F and P represent the dimensions of the fovea and periphery, respectively. For example, $[20, 30]$ corresponds to a 20×20 fovea and a 30×30 periphery. When $P = 0$, the agent receives input only from its fovea. In the original *SUGARL* implementation, the input is 84×84 pixels and contains the downsampled periphery with the high-resolution fovea overlayed. We also investigate efficiency with input separation as will be discussed later.

To ensure consistency across experiments and reduce the dimensionality of the design space, we fixed several key parameters:

- **Agent memory.** We used a simple frame-stacking approach to provide temporal context: the last four observations were concatenated along the channel dimension. This method is parameter-free (aside from the number of frames) and commonly used in vision-based RL. However, as we will discuss, this might pose limitations on the agent's ability to remember information across time.
- **Sensory action space.** The fovea's position was selected from an *absolute* action space: the agent could move the fovea from any location to any other location directly, rather than relying on gradual shifts. The action space was discretized into a 4×4 grid, allowing for 16 possible foveal positions. Provided the fovea size is smaller than one-quarter of the environment's width/height, this setup allows full coverage of the visual field.
- **RL algorithm.** We used the DQN algorithm with standard hyperparameters: `total_timesteps` = 10M, `learning_rate` = 0.0001, `target_network_update_freq` = 1000, among others. For a complete list of configuration parameters, we refer the reader to our codebase.

D. SUGARL

The core of our approach is based on the SUGARL framework [1]. The reader is encouraged to consult the original paper for more details, but we briefly summarize the key ideas here.

SUGARL introduces an intrinsic reward to guide the sensory policy, which determines where the agent should focus its fovea. This intrinsic reward complements the external environment reward and encourages informative perception.

The intrinsic reward is based on the predictability of the action a_t given the observations before and after it, i.e., the probability $p(a_t \mid o_t, o_{t+1})$. This probability is estimated by a separate neural network. Intuitively, if a_t can be accurately predicted from o_t and o_{t+1}, it suggests that the sensory input was informative about the action's effect, and therefore the sensory policy is rewarded.

IV. Experiments and Results

A. Baseline

As a baseline for comparison, we trained a traditional full-vision agent without a fovea - $[0, 84]$ - using the same training framework as our other agents to ensure fairness. All code related to SUGARL and foveated inputs was removed. The training rewards of this model serve as a reference point and are shown in Figure 2. The agent wins almost all its points after 2M steps, and then reaches an average score of 20.51.

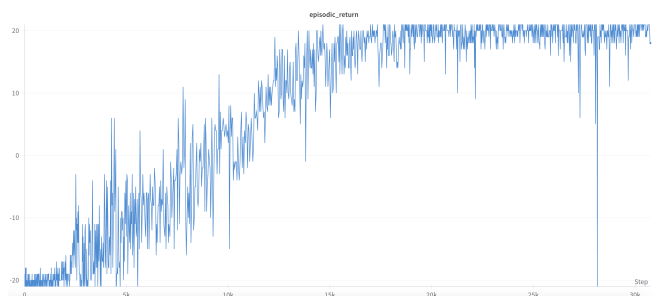


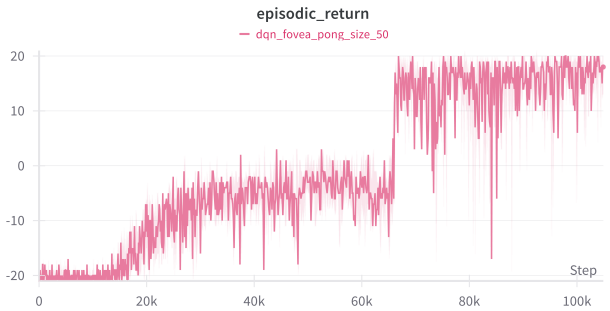
Figure 2: Full vision baseline agent

B. Fovea only

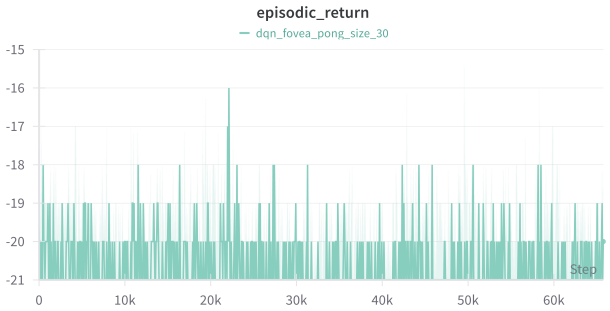
We start by investigating set-ups with fovea only, and no blurry periphery $[F, 0]$. We aim to answer the question: what is the smallest fovea size that still allows the model to learn and score well. After multiple experiments (including $[40, 0]$) we find that the minimum size that allows an agent to effectively learn is $[50, 0]$ 3 with an average reward of 19.33 IV.

C. Fovea + Peripheral

We extended the approach by combining the fovea with a low-resolution peripheral view. Using a [21, 30] setup, the model was able to learn effectively. However, video inspection showed that the fovea was not actively tracking the ball and instead moved seemingly at random. This suggests that the agent relied primarily on the peripheral input, which still contained enough visual information to follow the ball. This highlights a limitation of our toy *Pong* setup: the game is simple enough that the agent is not forced to optimize its sensory behavior.

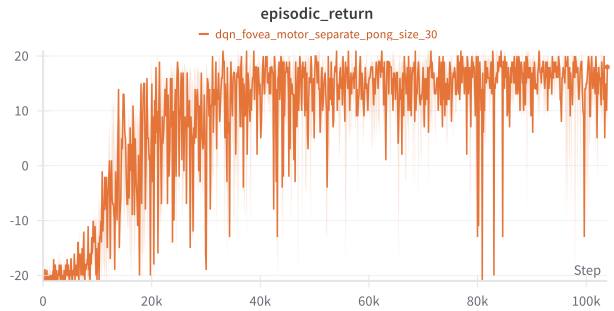


(a) Training reward [50, 0]

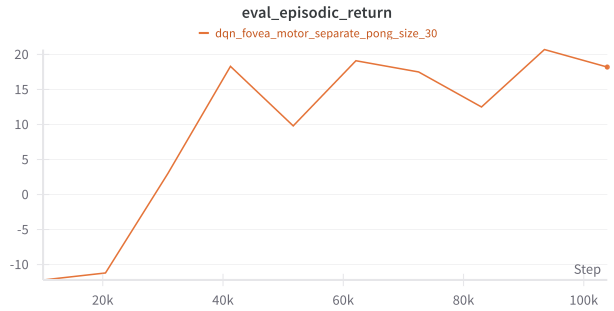


(b) Training reward fovea [30, 0]

Figure 3: Training breaks with fovea size below 50



(a) Training reward



(b) Evaluation reward during training

Figure 4: Evaluation and training reward for [30, 0] with computer vision tracking of the ball. We can see that this method learns almost as fast as our full vision baseline 2, meaning that tracking the ball is a good heuristic for this type of game.

To test this hypothesis, we reduced the peripheral resolution to 20 pixels. The model failed to learn in this configuration, confirming that peripheral vision played a crucial role in performance.

To investigate the agent's inability to learn with a [30, 20] setup, we manually designed a sensory policy that actively kept the ball centered in the fovea (using computer vision). When the ball was no longer visible, the policy defaulted to fixating on the lower right corner. Using this sensory strategy, the agent was able to learn effectively 4. This suggests that the limitation lies in the learning process rather than in the possibility of playing well under this visual constraint.

D. LCA

LCA (local competitive algorithm) [3] is a method developed to produce sparse representations in neural networks, originally inspired by how neurons in the brain compete and cooperate to represent stimuli efficiently.

We explored LCA as a mechanism to reduce computational cost at inference time by inducing sparsity via pruning. However, the results were inconclusive. As shown in Figure 5, the model exhibits high variance in performance even without sparsity constraints. While the mean reward remains relatively stable under pruning, this pattern is also observed in the fovea-only motor-separate baseline. This suggests that maybe the original model was already well-

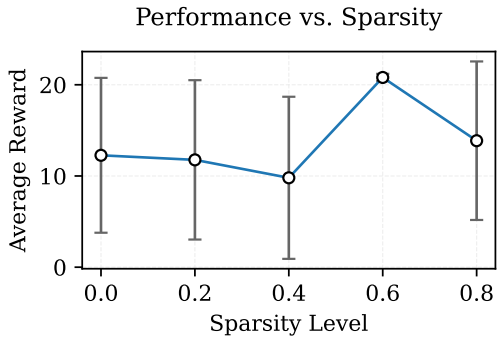


Figure 5: Comparing sparsity performance for LCA on [30, 0] with computer vision tracking of the ball.

suited to achieve sparsity, because for the Fovea-only input, the background is entirely black. It only helps the model infer its position, as it is not possible in this environment to obtain gradient flow from the fovea alone.

E. Efficiency

Standard Gym environments typically provide observations as 84×84 pixel images. In the original *SUGARL*

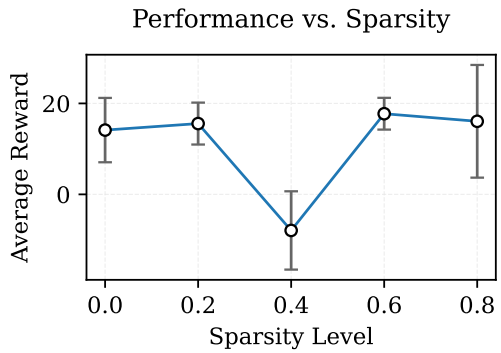


Figure 6: Comparing sparsity performance without LCA on [30, 0] with computer vision tracking of the ball.

implementation, although the input was downsampled, the full 84×84 frame was still used, with a high-resolution fovea overlaid on a blurred peripheral background.

In this work, we explore a fully separated input representation: a dedicated foveal patch of size $F \times F$, a peripheral view of size $P \times P$, and the fovea's spatial coordinates as a 1×2 vector. This decomposition significantly reduces the input dimensionality. For example, with [20, 30], the total number of input pixels drops by a factor of approximately 5.4 compared to the full 84×84 image. Given that the computational cost of convolutional layers scales linearly with the number of input pixels[1], this reduction suggests the potential for substantial improvements in inference speed.

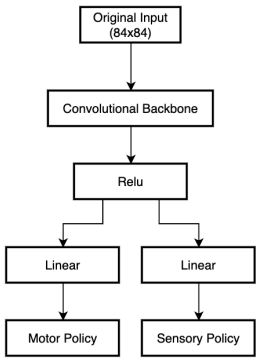


Figure 7: Initial architecture

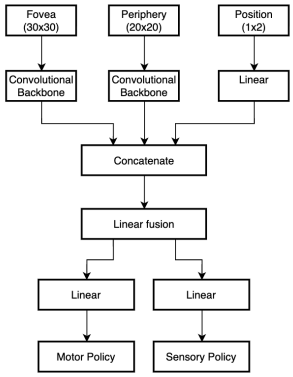


Figure 8: Our modified architecture

However, the situation is more nuanced. While the input size is reduced, our architecture processes each modality—fovea, periphery, and position—through separate convolutional or fully connected towers. This modular design introduces additional layers and parameters, which can offset the benefits of reduced input resolution. Furthermore, convolutions are parallelizable. As a result, the non-parallelized processing of separate convolution towers may lead to slower inference despite the smaller input size.

To better understand this trade-off between architectural modularity and computational efficiency, we report multiple metrics across different models: average episodic reward after training, average inference time per frame, total number of operations (FLOPs) per inference pass, and total number of trainable parameters.

We compare 3 models. The original SUGARL model (A) [1] and our two custom ones B and C (where C has larger convolution towers than B). Results are shown in I II III.

Table I: [30, 0]

Model	Avg Reward	Parameters (M)	Inference Time (ms)	Operations (MFLOP)
A	-20.93	1.69	0.69	6.90
B	-20.97	1.86	0.55	3.02
C	-20.94	3.76	1.19	16.4

Table II: [50, 0]

Model	Avg Reward	Parameters (M)	Inference Time (ms)	Operations (MFLOP)
A	19.33	1.69	0.69	6.90
B	-6.46	5.14	1.16	8.37
C	6.18	10.31	3.26	45.5

Table III: [21, 30]

Model	Avg Reward	Parameters (M)	Inference Time (ms)	Operations (MFLOP)
A	20.46	1.69	0.69	6.90
B	-5.88	2.85	0.69	3.53
C	20.21	9.43	2.70	20.5

As the tables show, the smaller separated model often reduces the total number of floating-point operations, indicating improved computational efficiency. However, it doesn't reach good scores. One reason may be the model's reduced capacity, but also potential redundancy in its computations—for instance, parts of the foveal region are processed both in the fovea and the peripheral pathways. The larger separated model does not improve efficiency— because both fovea and peripheral inputs remain relatively large—but it does perform comparably to the original full-input model in some instances.

Exploring more intermediate model sizes (between B and C) together with other potential improvements could allow us to find a model that performs well while being efficient.

F. Summarized results

As shown in Table IV, the SugaRL method achieves strong performance across a wide range of conditions. However, its effectiveness diminishes when the fovea size is less than 50 in the fovea-only setting, or when it is greater than or equal to 30 in some configurations. These results

[1]Convolutional complexity is $\mathcal{O}(C_{\text{in}} \cdot K^2 \cdot H \cdot W)$ where C_{in} is the number of input channels, K the kernel size, and $H \times W$ the spatial dimensions.

highlight a limitation of the approach: with small fovea sizes, the model struggles to learn effectively. Nevertheless, we demonstrate that good performance is still achievable with a fovea size as small as 30 by using a simple heuristic that focuses on tracking the ball. This indicates that models can successfully perform diverse tasks even with a significantly reduced field of view (FOV).

Table IV: Average Rewards of Different Methods

Method	FOV Size	Peripheral Res	Avg Reward	Std Reward
fovea_cv_tracking_pong_lca	30	None	12.27	8.49
fovea_cv_tracking_pong	30	None	13.93	6.84
fovea_peripheral_cv_tracking_pong	21	20	19.10	3.68
fovea_peripheral_pong (21 FOV)	21	30	20.47	0.50
fovea_pong (50 FOV)	50	None	19.33	0.47
fovea_peripheral_pong (30 FOV)	30	20	21.00	0.00
fovea_peripheral_pong (50 FOV)	50	20	20.77	0.42
fovea_pong (30 FOV)	30	None	-20.93	0.36
full_vision	None	None	20.51	2.30

V. Conclusion and Limitations

In this project we investigated the efficacy of foveated vision, compared between different foveated methods against a full vision baseline and investigated what type of performance gains one could have from using a fovea with peripheral vision or just a fovea.

Key findings demonstrate that SUGARL agents can effectively learn to play Pong but that the method is not perfect. Performance was strong with a fovea-only setup using a 50x50 pixel fovea, or when combining a smaller fovea (e.g., 21x21 pixels) with a low-resolution peripheral view (e.g., 30x30 pixels). Experiments highlighted that when both fovea and periphery are present, the agent can tend to rely significantly on the peripheral input, and performance degrades if peripheral resolution is too low and fovea size is to small. A significant limitation of the SUGARL method was observed when using very small fovea sizes (e.g., 30x30 pixels fovea-only), where the model failed to learn. However, a handcrafted vision policy (ball-tracking) combined with a separately trained motor policy demonstrated that learning is possible even with such a small fovea, suggesting the limitation lies more with SUGARL's learning capacity under heavier visual constraints rather than a model being able to learn with less visual information.

The project also explored an input resolution optimization technique by decoupling the foveal and peripheral visual streams and providing fovea coordinates explicitly. This approach theoretically offers a substantial reduction in input pixel count. The smaller custom architecture failed to yield any meaningful results, while model C achieved competitive performance in some cases (20.21 avg reward with 21px fovea and 30px periphery) comparable to the baseline SUGARL (Model A, 20.46 avg reward), without translating into superior computational efficiency in terms of total parameters, inference time, or FLOPs in this specific implementation. For instance, Model C had more parameters (9.43M vs 1.69M), slower inference (2.70ms vs 0.69ms), and more operations (20.5 MFLO vs 6.90 MFLO) than the baseline Model A for this configuration. This indicates that while reducing input data is a valid strategy, the architectural choices for processing these separate streams can introduce overheads that offset these gains. Experiments with LCA for inducing sparsity were inconclusive.

Future work could also explore more biologically plausible foveas — for instance, circular ones with gradually decreasing resolution toward the periphery, better mimicking the human visual system.

A. Main limitation and improvement possibility

In our opinion, the limited environment knowledge available at a given timestep due to foveated vision imposes a strong requirement on the agent's memory capabilities. To reduce the fovea size and peripheral resolution, an agent must be able to remember many features of the environment without observing them for several timesteps. Our current approach—stacking the last 4 frames—is by far the weakest link in the chain (and it is highly unefficient). We believe that incorporating more advanced mechanisms, such as LSTMs with a latent representation of previous states (i.e., memory not limited to a fixed number of steps), should be a central focus of any future project aiming to demonstrate efficiency improvements with SUGARL.

VI. Individual Contributions

J.S. A.N. and M.B. co-authored the report.

A.N. and J.S. adapted the SUGARL codebase and conducted the initial foveated training experiments.

A.N. further improved the training procedure to obtain the first promising results and, implemented the benchmark agents (no vision and together with **J.S.**, traditional full-vision).

A.N. developed the computer vision-based ball tracking system and the model where the motor policy is trained separately.

A.N. and J.S. developed the fovea visualization code.

J.S. and A.N. also conducted preliminary background research.

M.B. designed the model with separated periphery and fovea, with some preliminary work by **J.S.**.

A.N. explored the application of LCA (Locally Competitive Algorithm) to this problem.

J.S. investigated the computational efficiency of the different models.

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